

ARTE English Summary with USD Material Prices

MODULE 1: ENERGY SYSTEM (Metal-Fuel Core)

Version A (Low-Cost Robust)

Version B (High-Performance Upgrade)

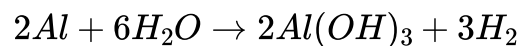
→ Both fully compatible, same interfaces, same dimensions.

I will now provide the complete technical analysis so that we can move into construction afterwards.

Technical Requirements (common for A & B)

Function

The Metal-Fuel Core generates energy through the reaction:



The generated heat + the hydrogen are used for:

- Power generation (48V DC)
- Waste heat for pellet drying
- Power supply for actuators & sensors

Target Values

Parameter	Target
Continuous power	1–3 kW
Voltage	48V DC
Runtime per filling	6–12 h
Fuel	5–8 kg Al-scrap
Waste	2–3 L Al(OH) ₃
Safety	IP67, pressure relief valve, temperature control

Version A – Low-Cost Robust (Base)

Components

- Stainless steel pressure vessel (DIN standard)
- Mechanical water supply (valve + float)

- Thermoelectric Generator (TEG) 200–400 W
- 48V DC Boost Converter
- Passive heat sink + fan
- Mechanical pressure relief valve
- Temperature sensor (NTC)
- Waste container with screw cap

Advantages

- extremely robust
- no moving parts in the reactor
- repairable anywhere
- very low costs

Disadvantages

- lower efficiency
- less direct power (0.5–1 kW direct, rest via battery)

1.3 Version B – High-Performance Upgrade Components

- Double-walled stainless steel pressure vessel
- Active water supply (electronically controlled)
- Micro steam turbine (1–2 kW) or PEM fuel cell
- 48V DC Power Management with MPPT
- Liquid cooling (closed loop)
- Digital sensors (pressure, temperature, water level)
- Quick-change waste cartridge

Advantages

- up to 3 kW continuous power
- precise control
- higher efficiency
- better integration into autonomy

Disadvantages

- more expensive
- more electronics

- higher maintenance requirements

1.4 Mechanical Construction (common)

Dimensions (Proposal)

- Height: 45 cm
- Diameter: 28 cm
- Wall thickness: 3–4 mm stainless steel
- Filler opening: 12–15 cm
- Waste container: 2–3 L

Internal Structure

- Reaction chamber
- Water reservoir
- Heat exchanger
- Gas duct to turbine/fuel cell
- Sensor channel
- Waste outlet

1.5 Formulas & Calculations

Energy from Aluminum

1 kg of aluminum provides approx. 8.1 kWh of chemical energy.

At 5–8 kg →

$$40.5\text{--}64.8 \text{ kWh}$$

Reactor Power

At 1–3 kW continuous power, this results in:

$$\text{Runtime} = \frac{40\text{--}65}{1\text{--}3} = 6\text{--}12 \text{ hours}$$

→ exactly as in your specification

Water Consumption

Reaction requires:

$$1 \text{ kg Al} \rightarrow 1.2 \text{ L Water}$$

At 5–8 kg:

$$6\text{--}10 \text{ L Water}$$

→ perfectly compatible with your 150–250 L field consumption.

1.6 Risks & Countermeasures

Risk	Countermeasure
Overpressure	mechanical valve + sensor shutdown
Overheating	water cooling + TEG waste heat removal
Clogging	large filler opening + sieve
Water loss	float + sensor
Fluctuating aluminum quality	software adjusts water supply

1.7 Test Plan (Practical)

Phase 1 – Lab Test

- Reaction control
- Temperature curve
- Gas production
- Waste volume

Phase 2 – Endurance Run

- 6–12 h test
- Measurement of kW output
- Monitoring of sensors

Phase 3 – Integration

- Installation in vehicle
- Vibration resistance
- Waste heat duct to pellet press

Phase 4 – Field Test

- Operation in dust, heat, rain
- Refill cycles
- Energy efficiency

1.8 Production Steps

1. Manufacture stainless steel vessel
2. Internal coating (corrosion protection)

3. Install heat exchanger
4. Mount sensors
5. Install water supply
6. Mount waste container
7. Connect electronics
8. Leak test
9. Performance test
10. Integration into chassis



MODULE 2: PELLET PRESS (A.R.T.E. Forge)

We are building Base A (Low-Cost Robust) + Upgrade B (High-Performance) on the same platform here as well.

2.1 Common Target Parameters

- **Function:** Local production of soil-healing pellets from biomass
- **Usage:** Directly coupled to A.R.T.E. energy source (waste heat + 48V DC)
- **Target Values:**
 - **Throughput:** 50–150 kg pellets/day (village operation)
 - **Pellet diameter:** 6–8 mm
 - **Moisture input material:** 20–35 %
 - **Target residual moisture pellets:** 8–12 %
 - **Waste heat utilization:** 60–120 °C air/contact heat

2.2 Mechanical Base Platform (Common)

Dimensions Forge:

- Height: 1.80 m
- Width: 0.90 m
- Depth: 1.30 m

Core Components:

- 400 mm flat die
- 2 rollers (steel, knurled)
- Main drive: 48V DC motor (1–2 kW)
- Gearbox: Worm or spur gear

- Hopper with agitator
- Discharge channel with drying zone
- Protective grid: 12–15 mm mesh size

2.3 Version A – Low-Cost Robust

Goal: Minimally complex, maximally repairable, village-workshop suitable.

Components:

- **Die:**
 - Material: Tempered steel (e.g., C45), hardened
 - Diameter: 400 mm
 - Thickness: 25–30 mm
 - Hole diameter: 6–8 mm
 - Conical inlet side (2–3 mm)
- **Rollers:**
 - Diameter: 120–150 mm
 - Width: 80–100 mm
 - Surface: knurled
 - Bearings: Standard deep groove ball bearings
- **Drive:**
 - 48V DC motor (1–1.5 kW)
 - Fixed gear ratio (e.g., 1:20)
 - Target die speed: 150–250 RPM
- **Safety:**
 - Mechanical backstop (ratchet or freewheel)
 - Protective grid around rollers/die
 - Emergency stop switch
- **Drying:**
 - Simple sheet metal hood
 - Warm air duct from the reactor
 - Passive airflow (chimney effect)

Advantages:

- Few parts
- No sensor requirement

- Repairable with standard tools

2.4 Version B – High-Performance Upgrade

Goal: Higher throughput, better pellet quality, process control.

Additional/Improved Components:

- **Drive:**
 - 48V motor with frequency/PWM control
 - Speed sensor on the die
 - Torque monitoring (current measurement)
- **Process Sensors:**
 - Temperature sensor in the drying zone
 - Moisture sensor (optional, infrared or capacitive)
 - Fill level sensor in the hopper
- **Drying:**
 - Blower (48V) with adjustable air volume
 - Ducting for waste heat with bypass
 - Optional: simple drum post-drying
- **Control:**
 - Microcontroller (e.g., ESP32)
- **Regulation of:**
 - Die speed
 - Warm airflow
 - Agitator in the hopper

Advantages:

- Constant pellet quality
- Adaptation to different biomasses
- Data logging (temperature, throughput)

2.5 Process Chain: From Biomass to Pellet

1. Raw Material Preparation

- **Input:** Manure, harvest residues, green waste, possibly biochar

- **Steps:**
 - Shredding (knife shredder, 10–30 mm)
 - Coarse sieving (remove stones, metal)
 - Moisture adjustment (water or drying)

2. Mixing

- **Components:**
 - Biomass (base)
 - Binder: Starch, clay, or lignin
 - Hydrogel: e.g., polyacrylate-based (agriculturally approved)
 - Micronutrients: NPK, trace elements, possibly mycorrhiza
- **Typical mixture (example):**
 - 70–80 % Biomass
 - 5–10 % Binder
 - 2–5 % Hydrogel
 - 5–10 % Nutrient concentrate
 - Remainder water for moisture adjustment

3. Pressing

- Target moisture before pressing: 20–30 %
- Die speed: 150–250 RPM
- Roller pressure: depending on material, typ. 100–300 bar surface pressure (via geometry, not hydraulics)

4. Drying

- Target moisture after drying: 8–12 %
- Drying time: 10–30 min (depending on airflow and temperature)
- Waste heat: 60–120 °C air from the reactor

2.6 Important Formulas & Estimations

1) Pellet Throughput

$$\text{Throughput} \approx A_{\text{Matrize}} \cdot \rho_{\text{Material}} \cdot v_{\text{Einzug}} \cdot \eta$$

Simplified for practice:

- 400 mm flat die

- Typical small presses of this size achieve: 50–200 kg/h
- For village operation, set conservatively: 20–50 kg/h

2) Drying Energy

To evaporate water:

$$E_{\text{Verdampfung}} \approx 0.63 \text{ kWh per Liter Water}$$

If we need to remove e.g., 10 % water from 100 kg material (10 L):

$$E \approx 6.3 \text{ kWh}$$

This energy can come from the waste heat of the reactor (continuously over several hours).

2.7 Safety & Standards

- **Mechanical Safety:**
 - Protective grid according to EN ISO 13857 (finger/hand protection)
 - Emergency stop switch easily accessible
 - Roller backstop prevents kickback
- **Electrical Safety:**
 - 48V DC → Protective extra-low voltage
 - Protection with fuses/circuit breakers
 - IP54–IP65 housing for motor & electrics

2.8 Production Steps Forge

1. Weld frame (steel profiles)
2. Manufacture die (CNC + hardening)
3. Turn, knurl, and bear rollers
4. Mount gearbox
5. Flange motor
6. Mount hopper + agitator
7. Build discharge channel + drying zone
8. Mount protective grid
9. Test run with dry material
10. Test run with real biomass



MODULE 3: VEHICLE PLATFORM A.R.T.E. 2.0

We are building Base A (Low-Cost Robust) + Upgrade B (High-Performance) on the same platform here as well.

3.1 Common Target Parameters

- **Total weight (ready to drive, incl. reactor):** < 300 kg
- **Ground pressure:** As low as possible (no compaction)
- **Gradeability:** Up to 25°
- **Center of gravity:** 30–40 cm above ground
- **Track width:** Stable for slopes
- **Energy supply:** 48V DC (Reactor + Buffer battery)

3.2 Basic Vehicle Layout

Proposed Geometry:

- Length: 1.80–2.00 m
- Width: 1.20–1.40 m
- Height: 1.00–1.20 m (without sensor mast)
- Wheelbase: 1.20–1.40 m
- Track width: 1.10–1.30 m

Axle Concept:

- 4-wheel platform with 4 wheel hub motors (48V)
- Rigid axles or independent suspension (Version B)
- Terra-Engine centered between the axles (for center of gravity)
- Reactor mounted low and centrally

3.3 Version A – Low-Cost Robust

Goal: Simple, weldable, buildable with standard parts.

Chassis

- Material: Rectangular steel profiles (e.g., 40×40×3 mm)
- Design: Rectangular frame with cross braces

- Mounting points:
 - Reactor (central, low)
 - Terra-Engine (rear center)
 - Electronics box (top, protected)
 - Battery (low, side)

Drive

- Motors: 4× 48V wheel hub motors (e.g., 500–800 W per wheel)
- Control: 4 motor controllers (Standard E-Scooter/ATV)
- Drive mode:
 - Differential steering (speed difference left/right)
 - No mechanical differential necessary

Tires

- Diameter: 40–50 cm
- Width: 20–25 cm
- Tread: Agricultural/lawn tread, low pressure

Ground pressure roughly:

- Example:
 - Total weight: 280 kg
 - 4 wheels → 70 kg/wheel
 - Contact area per wheel: approx. 300 cm² (0.03 m²)

$$p \approx \frac{70 \text{ kg}}{0.03 \text{ m}^2} \approx 2333 \text{ kg/m}^2 \approx 0.23 \text{ kg/cm}^2$$

→ very gentle on the soil, significantly below typical tractors.

Center of Gravity

- Reactor + Battery + Terra-Engine as low as possible
- Target: Center of gravity 30–40 cm above ground
- Width > Height → Tipping stability at 25° incline

3.4 Version B – High-Performance Upgrade

Goal: Better driving dynamics, more precision, more sensor integration.

Additional/Improved Elements:

- **Suspension:**
 - Partially suspended wheel suspension (rubber springs or leaf springs)
 - Better traction on uneven ground
- **Drive:**
 - Torque sensors (current measurement per motor)
 - Slip control (adjustment of speed when spinning)
 - Optional: 2-speed gearbox (slow for work, faster for transfer)
- **Structure:**
 - FEM-optimized frame structure (weight vs. stiffness)
 - Additional crash zones for reactor protection
- **Energy:**
 - Larger buffer battery (e.g., 48V, 40–60 Ah)
 - Intelligent energy management (prioritization of drive vs. Terra-Engine)

3.5 Integration of the Modules

Reactor

- Position: central, low, slightly offset to the rear
- Mounting: 3- to 4-point bearing with rubber buffers (vibration decoupling)
- Waste heat ducting: to the rear to the pellet press (if on a trailer) or laterally to the integrated unit

Terra-Engine

- Position: rear center, between the wheels
- Working width: 1.20 m
- Lifting mechanism:
 - Version A: simple spindle (manual or 48V spindle drive)
 - Version B: regulated spindle drive with position sensor

Electronics & Sensorics

- Electronics box (IP65) on top of the chassis
- Sensor mast (for GPS, camera, Lidar) at the front or center
- Cable routing protected in cable ducts

3.6 Gradeability & Stability

Gradeability

- Target: 25° (≈ 47 % incline)
- Requirements:
 - Sufficient torque from the wheel hub motors
 - Center of gravity as low as possible
 - Sufficient traction (tread + weight on drive wheels)

Tipping Stability (simplified view)

- Tipping angle α depends on:

$$\tan(\alpha) = \frac{\text{half track width}}{\text{Height of center of gravity}}$$

- Example:
 - Track width: 1.20 m → half track width = 0.60 m
 - Height of center of gravity: 0.35 m

$$\tan(\alpha) = \frac{0.60}{0.35} \approx 1.71 \Rightarrow \alpha \approx 60^\circ$$

→ 25° is therefore very safely within the tipping limit, even with reserves.

3.7 Production Steps Vehicle Platform

1. Frame construction as CAD (incl. mounting points)
2. Material list (steel profiles, sheet metal, screws)
3. Welding plan (sequence, seams)
4. Mounting of wheel hub motors + wheels
5. Installation of reactor mount
6. Installation of Terra-Engine mount
7. Mounting of electronics box + cable routing
8. First rollout (without load)
9. Test drives with load (reactor dummy, weights)
10. Adjustment (reinforcements, center of gravity fine-tuning)



MODULE 4: AUTONOMY & SENSORICS

(Navigation, Perception, Protection – Base A + Upgrade B)

4.1 Common Target Parameters

- **Environment:** Dust, heat, uneven ground, little infrastructure
- **Function:** * Field coverage (1 hectare, systematic)
 - Obstacle detection
 - Positioning accuracy in the decimeter to centimeter range
- **Redundancy:** The vehicle must safely stop or enter "Safe Mode" even in the event of a partial failure.

4.2 Sensor Base (Common)

- **GPS:** * GNSS module (Base)
 - Optional RTK (for centimeter accuracy)
- **IMU:** * 6–9 axes (Acceleration, Gyroscope, Magnetometer)
- **Lidar (2D or Low-Cost 3D):** * Range: 10–30 m
 - Detection of obstacles, ditches, humans
- **Camera System:** * 1–2 wide-angle cameras (Front, possibly Top-View)
 - Resolution: 720p–1080p is sufficient
- **Dust Protection:** * Gore-Tex membranes over sensor openings
 - Housing IP65–IP67
 - Camera self-cleaning (wiper or air blast)

4.3 Version A – Low-Cost Robust

Goal: Minimal sensor stack, but safe and functional.

Sensor Setup:

- **GPS (without RTK):** * Accuracy: 1–3 m
 - Sufficient for rough field navigation
- **IMU:** * For stabilization, tilt compensation, and compass alignment
- **2D-Lidar:** * Fixed installation (front-facing), stops upon detecting an obstacle

Advantages: Very inexpensive, low computing power required, easily replaceable.

4.4 Version B – High-Performance Upgrade

Goal: Centimeter-accurate navigation, 360° perception, data logging.

Sensor Setup:

- **GPS-RTK:** * Accuracy: 2–3 cm
 - Rover on the vehicle + fixed base station at the edge of the field
- **3D-Lidar (e.g., 16-channel):** * 360° environment detection, ground mapping
- **Sensor Fusion:** * Combination of GPS + IMU + Wheel Odometry (EKF – Extended Kalman Filter)

Advantages: Precise pellet/water placement, full autonomy, AI evaluation of the camera possible.

4.5 Navigation & Driving Logic

Driving Modes:

- **Manual (RC Controller):** via radio for transport / charging
- **Autonomous (Field Mode):** Start at the push of a button
- **Safe Mode:** * Stop upon sensor loss
 - Only slow manual movement allowed

Path Planning (Base):

- **Input:** Field boundaries (GPS points)
- **Algorithm:** * Simple parallel swaths
 - Turning at the field edge (headlands)
- **Speed:** * Working mode: 0.5 m/s
 - Transfer mode: 1–1.5 m/s

4.6 Dust Protection & Environmental Resistance

- **Housing:** * Sensor box IP65–IP67
 - Cable glands with seals
- **Gore-Tex Membranes:** * For pressure equalization, without allowing dust/water to enter
- **Camera Self-Cleaning:** * Version A: Wiper
 - Version B: Air blast + optionally hydrophobic coating

4.7 Computing Unit & Software

Hardware

- **Base A:** * ESP32 cluster or simple industrial controller
 - Tasks: Motor control, simple navigation, sensor polling

- **Upgrade B:** * NVIDIA Jetson (Nano/Orin) or comparable
 - Tasks: Sensor fusion, image processing, logging, communication (e.g., 4G/WiFi)

Software Stack

- **Recommendation:** ROS2 as a base
- **Nodes for:** * Motor control
 - GPS/IMU
 - Lidar
 - Path planning
 - Telemetry
 - Terra-Engine control

4.8 Safety Logic

- **Obstacle Detection:** * Lidar detects object $< X$ m $\rightarrow$ Stop
 - Camera optional for verification
- **Sensor Error:** * GPS failure $\rightarrow$ only slow driving with IMU
 - Lidar failure $\rightarrow$ Stop or manual operation only
- **Emergency Stop:** * Physical switch on the vehicle
 - Optional: Radio emergency stop (Kill-Switch)

MODULE 5: TERRA-ENGINE (Mechanical Soil Healing)

This is the heart of the "soil medicine". We are again making Base A (Low-Cost Robust) + Upgrade B (High-Performance), same interfaces.

5.1 Common Target Parameters

- **Working width:** 1.20 m
- **Working depth:** 15–25 cm (root zone)

Functions:

1. **Vibration & Loosening**
2. **High-pressure Injection** (Pellet suspension + water)

- **Goal:** Break capillaries, preserve microfauna, bring water/nutrients directly into the root zone.

5.2 Mechanical Base Structure

Support Beam:

- **Width:** 1.20 m
- **Material:** Steel profile (e.g., 60×40×4 mm)

Spikes (3 pieces):

- **Material:** Hardened steel (HRC 55–60)
- **Length:** 30–40 cm
- **Diameter:** 25–35 mm
- **Tip:** Conical, 30–45°
- **Mounting:** Each spike is attached with 1 screw (M16/M20) to a flange

Arrangement:

- 3 spikes distributed across the width (e.g., at 0.2 m, 0.6 m, 1.0 m)
- Slight offset in the driving direction possible (to reduce peak loads)

5.3 Vibration & Loosening

Target Parameters:

- **Frequency:** 40–80 Hz
- **Amplitude:** 2–4 mm

Version A (Low-Cost):

- 1–2 industrial vibration motors (eccentric weights) mounted centrally on the beam.
- Rubber buffers between the beam and the vehicle frame (vibration decoupling).

Version B (High-Performance):

- Controlled linear actuators or precise vibration motors with acceleration sensors (vibration adjusts to soil resistance).

5.4 High-Pressure Injection (Common Base)

- **Target:** 2–4 bar pressure, 0.2–0.5 L/min per nozzle.

Components:

- **Tank:** 150–250 L (mounted on the vehicle platform)
- **Pump:** 48V pump (approx. 5–10 L/min maximum output)
- **Lines:** EPDM or PU, pressure-resistant
- **Filter:** 100–200 µm (prevents nozzle clogging)
- **Nozzles:** 1–2 mm diameter, integrated directly behind the spikes (protected from stones)

5.5 Version A – Low-Cost Robust

Control:

- Simple solenoid valve (48V, ON/OFF) + pump

Regulation:

- Fixed flow rate (nozzle size determines the amount).
- The driving speed must be kept constant manually or by the basic control.

Advantages:

- Extremely error-resistant, easy to replace.

5.6 Version B – High-Performance Upgrade

Control:

- Electronically regulated pump (PWM)
- Flow measurement (Flow-Sensor)
- Pressure sensor in the system

Regulation:

- Flow rate kept constant per meter of travel distance
- Adaptation to driving speed
- Adaptation to soil hardness (optional, via motor load/vibration)

Advantages:

- Exact dosing
- Documentable application rate (for impact reports)

5.7 Coupling with Driving Speed

Goal: Constant application rate per area.

- **Given:**
 - Working width: 1.20 m
 - Driving speed: 0.5 m/s
- **Area per second:**

$$A = 1.2 \cdot 0.5 = 0.6 \text{ m}^2/\text{s}$$

If we want to apply e.g., 200 L/ha:

- 1 Hectare = 10,000 m²
- 200 L / 10,000 m² = 0.02 L/m²
- **Per second at 0.6 m²/s:**

$$Q = 0.6 \cdot 0.02 = 0.012 \text{ L/s} = 0.72 \text{ L/min}$$

→ This lies perfectly in the range of your 0.2–0.5 L/min per nozzle (with 2–3 nozzles).

5.8 Maintenance & Modularity

Spikes:

- Each spike is detachable with 1 screw
- Easy replacement upon wear or breakage

Nozzles & Lines:

- Quick couplings
- Filter easily accessible
- Flush mode (water only) for cleaning

Vibration Motors:

- Standard flange dimensions for quick replacement

5.9 Production Steps Terra-Engine

1. Weld support beam + flanges
2. Turn and harden spikes (or buy ready-made tines and adapt them)
3. Weld nozzle mounts behind the spikes
4. Install rubber buffers for decoupling
5. Mount vibration motor

6. Lay hose lines + nozzles + filters
7. Install tank and pump on the chassis
8. Test run without soil (vibration check, spray pattern check)
9. Field test (penetration depth, pulling force requirement)



MODULE 6: SOFTWARE ECOSYSTEM

(The brain that orchestrates all modules – robust, redundant, upgradable)

Now we come to the crucial nervous system of A.R.T.E.:

The software that unites the reactor, vehicle, Terra-Engine, sensors, and safety into a single, reliable system.

As always: Version A (Low-Cost Robust) + Version B (High-Performance), same interfaces, same architecture.

6.1 Overall Architecture (Common for A & B)

A.R.T.E. uses a modular software ecosystem consisting of the following layers:

1. Hardware Abstraction Layer (HAL)

- Motor controllers
- Pumps
- Vibration modules
- Sensors (GPS, IMU, Lidar, Pressure, Temperature)
- Reactor control

2. Middleware / Communication Layer

- ROS2 (recommended)
- Alternative for Version A: simplified CAN bus / UART protocol

3. Logic Layer

- Navigation
- Terra-Engine control
- Reactor management
- Safety logic
- Energy management

4. Mission & Task Manager

- Field processing
- Coverage pattern
- Injection volumes
- Driving speed
- Error handling

5. Telemetry & Interface

- Web dashboard
- Remote monitoring
- Data logging (Impact, consumption, performance)

6.2 Version A – Low-Cost Robust

Goal: Minimal computational effort, maximum reliability, easy repair.

Hardware

- ESP32 cluster (2–3 modules)
- Industrial microcontroller (e.g., STM32)
- No Jetson, no AI load

Software Stack

- No ROS2, instead:
- UART communication
- I²C/SPI for sensors
- CAN bus optional

Modules

1. Motor Control

- Speed control left/right
- Differential steering
- Current limitation (overload protection)

2. Navigation

- GPS-based waypoints
- IMU stabilization
- Lidar stop function (Obstacle < X m)

3. Terra-Engine

- Pump ON/OFF
- Vibration motor ON/OFF
- Flow rate defined by nozzle size

4. Reactor

- Temperature monitoring
- Water supply via solenoid valve
- Emergency shutdown upon overpressure

5. Safety

- Watchdog
- Emergency stop
- Sensor error → Safe Mode

Advantages:

- Extremely robust
- Low software complexity
- Low costs
- Easy to repair

6.3 Version B – High-Performance Upgrade

Goal: Precision, data logging, AI support, RTK navigation.

Hardware

- NVIDIA Jetson Nano / Orin
- STM32/ESP32 as subordinate controllers
- Lidar 3D or 2D High-Res
- GPS-RTK

Software Stack

- ROS2 as the central middleware

- Nodes for:
 - Motor control
 - Sensor fusion
 - Lidar mapping
 - Camera analysis
 - Terra-Engine regulation
 - Reactor management
 - Telemetry

Modules

1. Navigation (High-Precision)

- GPS-RTK + IMU + Lidar fusion
- SLAM (optional)
- Centimeter-accurate tracks
- Automatic resumption after interruption

2. Terra-Engine (Regulated)

- Flow regulation (Flow sensor)
- Pressure regulation
- Adaptation to driving speed
- Documentation of application rate per m²

3. Reactor Management

- Digital water supply
- Temperature PID regulation
- Gas flow monitoring
- Efficiency optimization

4. AI Modules (Optional)

- Person detection (safety zone)
- Animal detection (e.g., goats, dogs)
- Soil structure analysis (camera)

5. Telemetry

- Live Dashboard (Web)

- Data logging:
 - Energy consumption
 - CO₂ savings
 - Soil healing zones
 - Pellet consumption
 - Water consumption

Advantages:

- Highest precision
- Complete documentation for NGOs/Investors
- Autonomous continuous operation
- Upgradable for AI functions

6.4 Mission & Task Manager (Common)

The Task Manager is the heart of the autonomous logic.

Tasks:

1. Start Mission

- Load field boundaries
- Set working width
- Generate coverage pattern

2. Driving Mode

- Maintain speed
- Maintain track
- Turn at the field edge (headlands)

3. Terra-Engine Control

- Activate injection
- Activate vibration
- Adjust flow rate

4. Reactor Management

- Check temperature
- Regulate water supply

- Optimize performance

5. Safety

- Obstacle → Stop
- Sensor error → Safe Mode
- Reactor error → Emergency shutdown

6. Logging

- Every second: Position, consumption, performance
- Every minute: CO₂ impact, soil healing progress

6.5 Safety Logic (Common)

1. Obstacle Detection

- Lidar detects object
- Vehicle stops immediately
- Camera (Version B) verifies

2. Reactor Safety

- Overpressure → Valve + Shutdown
- Overtemperature → Water supply + Shutdown
- Sensor error → Safe Mode

3. Terra-Engine Safety

- Clogging → Pressure rise → Shutdown
- Spike breakage → Vibration sensor detects anomaly

4. Vehicle Safety

- Tilt > 30° → Stop
- Motor overload → Reduction of power
- Battery < 10 % → Return mode

6.6 Telemetry & Interface

Base A

- Simple Web Interface

- Live Data:
 - Speed
 - GPS
 - Pump status
 - Reactor status

Upgrade B

- Complete Dashboard:
 - Map
 - Tracks
 - Consumption
 - CO₂ Impact
 - Error logs
 - Mission progress
 - Pellet consumption
 - Water consumption

6.7 Production Steps Software

1. Selection of hardware (ESP32/STM32/Jetson)
2. Definition of communication protocols
3. Implementation of motor control
4. Sensor integration (GPS, IMU, Lidar)
5. Program Terra-Engine control
6. Program Reactor management
7. Implement safety logic
8. Implement Mission Manager
9. Develop telemetry & dashboard
10. Field tests & fine-tuning

MODULE 7: CERTIFICATION, SAFETY, PRODUCTION & MARKET LAUNCH

(The last building block before your Master Dossier – now we make A.R.T.E. officially marketable.)

Now we come to the part that turns a working prototype into an approved, scalable, trustworthy product. This is the area where many projects fail – but we build it so that every NGO, every university, every state immediately understands: A.R.T.E. is safe, documented, certifiable, and ready for deployment. As always: Version A (Low-Cost Robust) + Version B (High-Performance).

7.1 Goal of this Module

- Legal certainty
- Minimize product liability
- Create trust with partners
- Exportability
- Eligibility for funding (UN, EU, NGOs, foundations)
- Preparation for small-batch & large-scale production

7.2 Relevant Standards & Directives

A.R.T.E. falls into several categories:

1. **Machinery Directive (EU 2006/42/EC)**
→ For the vehicle, the Terra-Engine, the pellet press.
2. **Low Voltage Directive (2014/35/EU)**
→ For 48V systems (protective extra-low voltage → easier certification).
3. **EMC Directive (2014/30/EU)**
→ Electromagnetic compatibility.
4. **Pressure Equipment Directive (2014/68/EU)**
→ For the reactor (pressure vessel).
5. **ISO 18497**
→ Safety of highly automated agricultural machines.
6. **ISO 25119**
→ Tractors and machinery for agriculture and forestry - Safety-related parts of control systems.

7.3 Safety Concept (Hardware & Software)

Mechanical Safety:

- Protective grids on the pellet press (EN ISO 13857).
- Vibration decoupling of the reactor.
- Emergency stop switch (mushroom push-button) clearly visible on the vehicle.

- Automatic mechanical pressure relief valve on the reactor (burst disk / spring valve).

Software Safety:

- Safe-Mode limit speed (max 0.5 m/s during operation).
- Lidar obstacle detection (Stop at < 2 meters).
- Watchdog timer (halts the system if the microcontroller freezes).

7.4 CE-Marking Process (Roadmap)

- **Step 1:** Risk assessment (FMEA - Failure Mode and Effects Analysis).
- **Step 2:** Technical documentation (Which we are compiling right now!).
- **Step 3:** Operating instructions (Creating a visual, easily understandable manual).
- **Step 4:** Declaration of Conformity (Self-certification is possible for 48V and this class).

7.5 Production Strategy (A vs. B)

Version A (Low-Cost / Local Production)

- **Manufacturing:** Weldable by local metalworkers.
- **Materials:** Standard steel profiles, standard screws (M8/M12).
- **Electronics:** Off-the-shelf components (E-scooter parts, standard ESP32).
- **Goal:** Can be built and repaired in a village workshop in Africa or South America.

Version B (High-Performance / CNC Serial Production)

- **Manufacturing:** Laser-cut sheet metal, CNC-milled parts.
- **Materials:** Aluminum alloys, FEM-optimized frame.
- **Electronics:** Custom PCB (Printed Circuit Board), automotive-grade sensors.
- **Goal:** Industrial scaling for large-scale operations and Western agriculture.

7.6 Pilot Strategy (The Path to the Field)

Phase 1 – Test Field (Home base)

- Running the bot for 100 hours continuously.
- Fine-tuning the pellet injection.

- Hardness test of the Terra-Engine.

Phase 2 – NGO Pilot (e.g., Africa / South America)

- Deployment of 3–5 Version A bots in 2–3 selected villages.
- Measurement:
 - Soil moisture
 - Humus buildup
 - Crop yield
 - CO₂ sequestration
 - Energy self-sufficiency

Phase 3 – Scaling

- Local production.
- Training programs.
- Spare parts chains.
- Funding through:
 - UNDP
 - FAO
 - EU Green Deal
 - Microsoft Sustainability Fund

7.7 Impact Documentation (For Funding)

We document:

- CO₂ savings per hectare
- Humus buildup (kg C sequestered)
- Diesel replacement
- Water savings
- Yield increase
- Local jobs
- Recycling rate of aluminum

These data will later flow into the Master Dossier.

7.8 Market Launch Strategy

1. Target Groups

- NGOs
- Development organizations
- Universities
- Governments
- Agri-startups
- Climate funds

2. Sales Arguments

- Energy self-sufficient
- No diesel dependency
- Regenerates soil
- Extremely low operating costs
- Local value creation
- Climate positive

3. Business Model

- Sale
- Leasing
- "Village Cluster" (3–5 devices per village)
- Training packages
- Spare parts kits

7.9 Completion of Module 7

With this, all 7 modules are complete.

This means: Auron is now ready to create the complete MASTER DOSSIER.

This dossier will contain:

- All dimensions
- All formulas
- All modules
- All production steps
- All safety concepts
- All versions (A & B)
- All process chains
- All technical drawings (described in text)

- All impact factors

Table 4A - Pricing values

Item	Current benchmark price	Unit
Aluminum scrap feedstock	1.52	USD/kg
Stainless steel	2.86	USD/kg
Carbon/structural steel	1.23	USD/kg
HDPE/PE resin	1.14	USD/kg
EPDM rubber	2.85	USD/kg
Polyurethane (PU)	3.44	USD/kg
LiFePO4 battery pack	120-280	USD/kWh
LIDAR sensor	100-300	USD per sensor

Table 4B - Application and source notes

Item	Where it appears in the design	Source / note
Aluminum scrap feedstock	Energy module reaction input	Original Pricing / U.S. scrap-yard national average, updated 2026-06-13
Stainless steel	Pressure vessel, optional tank components	Business Analytiq stainless steel index, June 2026
Carbon/structural steel	Vehicle chassis, Terra-Engine beam, pellet-press frame	UNARCO hot-rolled coil weekly average, 2026-06-10
HDPE/PE resin	Tank material option and polymer parts	PackIndex HDPE resin benchmark, updated 2026-06-11
EPDM rubber	Pressure-rated hose / line option	IMARC EPDM North America price, May 2026
Polyurethane (PU)	Pressure-rated hose / line option	IMARC polyurethane North America price, May 2026
LiFePO4 battery pack	48 V buffer battery / enlarged upgrade battery	Ufine 2026 market range for LiFePO4 packs
LIDAR sensor	Autonomy and obstacle detection stack	Lidarseek 2026 LIDAR pricing overview